AS · Unit 1 · WCH11 Structure, Bonding and Introduction to Organic Chemistry

Bonding and structure

Every chemical bond is an electrostatic attraction between positive and negative charges. What differs is where the charges are: between ions, between nuclei and shared electrons, or between metal ions and a sea of electrons. This topic explains each type of bond, the shapes that molecules take, how bonds become polar, and how the structure of a substance explains its melting temperature, conductivity and solubility.

SPEC 3.1 Evidence for the existence of ions

WHAT THIS SPEC POINT ASKS

Know and interpret the evidence that ions exist: the physical properties of ionic compounds, electron density maps, and the migration of ions.

Physical properties

Property of ionic compounds	What it tells you
High melting and boiling temperatures	strong forces act in every direction through a giant structure
Conduct electricity when molten or dissolved, but not when solid	charged particles exist; they are **fixed in place** in the solid but **free to move** in the liquid or solution
Often **soluble in water**, insoluble in non-polar solvents	charged particles are attracted to polar water molecules
Brittle: they shatter when hit	a small shift lines up like charges, which repel (3.4)
Electrolysis decomposes them, with products at both electrodes	positive particles go to the cathode, negative ones to the anode

Electron density maps

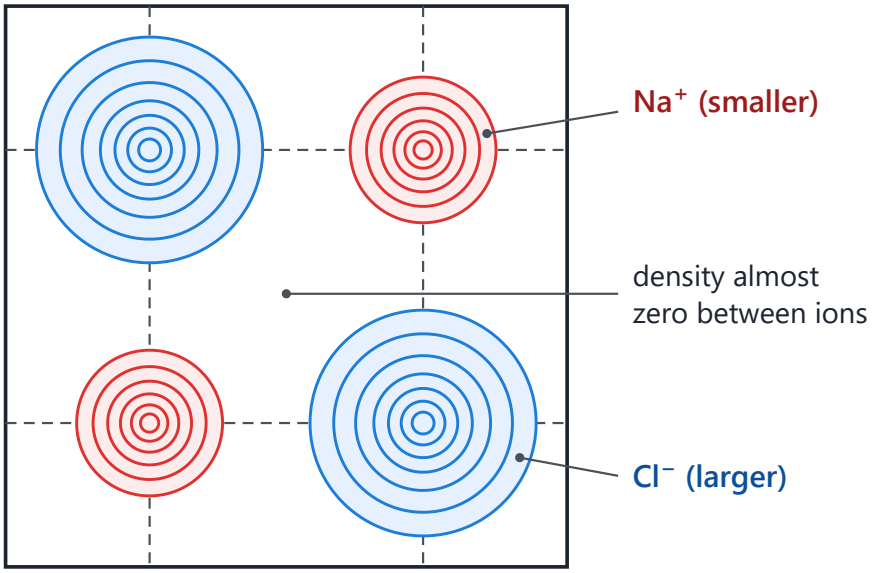
- X-ray diffraction measures how electron density is spread through a crystal. The result is drawn as a contour map, like a map of hills.
- In **sodium chloride** each ion has its **own separate set of roughly circular contours**, and the electron density falls to **almost zero between the ions**. There is no build-up of electrons between them, so electrons are not shared: they belong to one ion or the other.

- The Cl^- contours enclose a bigger region than the Na^+ contours: Cl^- is the larger ion. Compare this with a covalent bond (3.10), where contours **surround both nuclei** and density stays high between them.

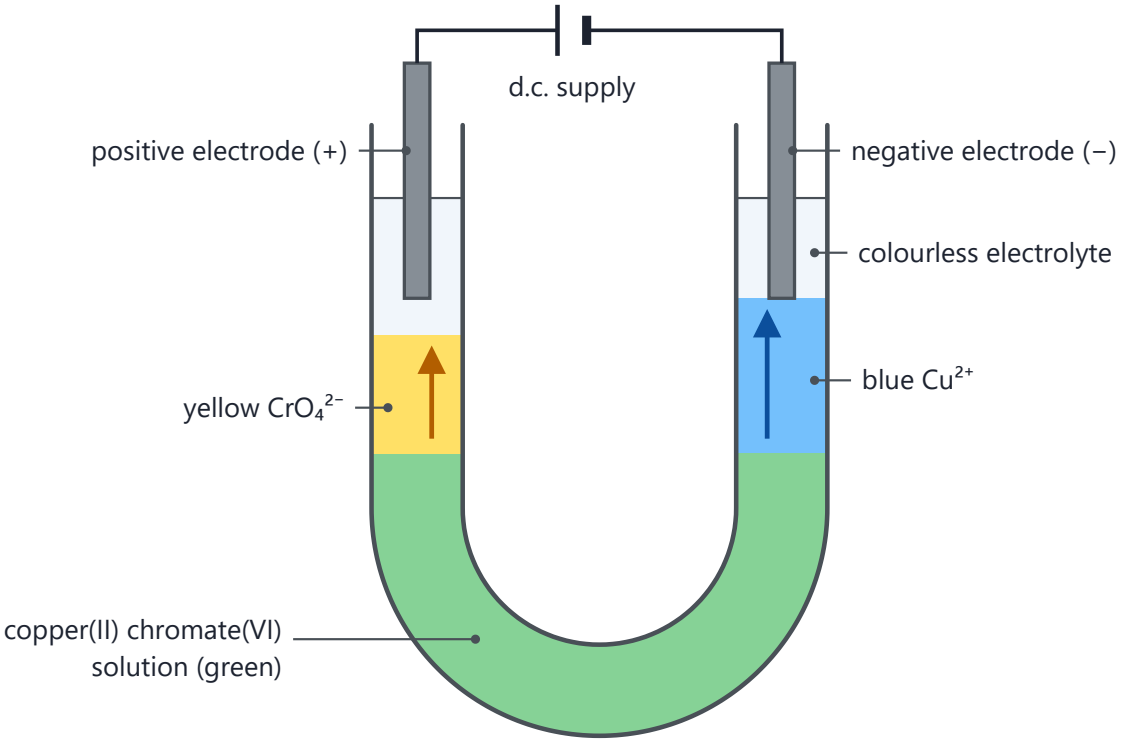
Migration of ions

If coloured ions move towards an electrode when a direct current flows, the colour must be carried by **charged** particles.

- **On a microscope slide**: wet a strip of filter paper with an electrolyte (e.g. sodium chloride or copper(II) sulfate solution), clip it to a slide, connect each end to a low-voltage d.c. supply, and put a crystal of **potassium manganate(VII)** in the middle. A **purple** colour moves towards the **positive electrode (anode)**: the MnO_4^- ion is negative. With a crystal of copper(II) sulfate, a **blue** colour moves towards the **negative electrode (cathode)**: Cu^{2+} is positive.
- **In a U-tube**: put green **copper(II) chromate(VI)** solution at the bottom and carefully layer a colourless electrolyte (e.g. dilute acid) on top in each arm. With a d.c. supply, a **blue** colour (Cu^{2+}) rises towards the **cathode** and a **yellow** colour (CrO_4^{2-}) towards the **anode**. The green colour was the two ions together.



An electron density map of sodium chloride. Each set of rings is one ion; the larger ones are Cl^- and the smaller ones Na^+.



Coloured ions move in opposite directions: blue Cu^{2+} to the negative electrode and yellow CrO_4^{2-} to the positive electrode.

MARK-SCHEME WORDING

- Migration (3 marks): electrolysis using a **d.c.** supply; a crystal or solution of copper(II) sulfate on **damp filter paper** (on a microscope slide), or a U-tube; "**blue** copper(II) ions move to the **negative** electrode / cathode". A labelled diagram is accepted.
- Solid ionic compound doesn't conduct because "the **ions** are fixed / not free to move". Not "there are no delocalised electrons".

Often asked as: describe an experiment to show ions migrate; MCQ on reading an electron density map; explain why an ionic compound conducts when molten but not when solid.

COMMON MISTAKE

Saying a solid ionic compound doesn't conduct because it has "no free electrons". It has **ions**; they just can't move. Saying the electrons move in a molten ionic compound: the **ions** move.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Colours as evidence: blue Cu^{2+} moves to the cathode, purple MnO_4^- to the anode (WCH11 Oct 2025).

SPEC 3.2 Forming ions by losing or gaining electrons

WHAT THIS SPEC POINT ASKS

Describe how ions form when atoms lose or gain electrons.

- **Metals** (few outer electrons) **lose** electrons to form **positive ions (cations)**. Na: 2,8,1 → Na^+: 2,8.
- **Non-metals** (nearly full outer shell) **gain** electrons to form **negative ions (anions)**. Cl: 2,8,7 → Cl^-: 2,8,8.
- In forming an ionic compound, electrons are **transferred** from the metal atoms to the non-metal atoms. The ions usually have the configuration of a **noble gas**.
- In s, p, d notation: Mg $1s^2\ 2s^2\ 2p^6\ 3s^2$ → Mg^{2+} $1s^2\ 2s^2\ 2p^6$ (two 3s electrons lost); O $1s^2\ 2s^2\ 2p^4$ → O^{2-} $1s^2\ 2s^2\ 2p^6$ (two electrons gained into 2p).
- The ion's charge follows from the group: Group 1 → 1+, Group 2 → 2+, Group 13 (Al) → 3+, Group 15 (N) → 3−, Group 16 → 2−, Group 17 → 1−.

MARK-SCHEME WORDING

- "Electrons are transferred from the metal to the non-metal"; give the number of electrons and the charges ("each Mg atom loses two electrons to form Mg^{2+}").

Often asked as: describe what happens to the electrons when sodium reacts with fluorine; part of a dot-and-cross question.

COMMON MISTAKE

Saying atoms "share" electrons in an ionic compound. Describing the ion as the atom ("magnesium has a 2+ charge"): say **magnesium ion**.

SPEC 3.3 Dot-and-cross diagrams for ionic compounds

WHAT THIS SPEC POINT ASKS

Draw dot-and-cross diagrams to show the electrons in cations and anions.

• electrons from the non-metal × electrons from the metal



sodium chloride, NaCl

magnesium oxide, MgO

Outer shells only. The cation's outer shell is shown empty here; drawing its full inner shell of 8 is also accepted.

How to draw one:

1. Draw **each ion separately**, in **square brackets**, with its **charge** at the top right.
2. Show the anion with a **full outer shell of 8**: its own electrons as dots, the electron(s) it gained as crosses.
3. Show the cation with **no outer electrons** (or with the full shell beneath).
4. Show the right **number** of each ion: $CaCl_2$ needs one Ca^{2+} and **two** Cl^- (or write "2" in front of $[Cl]^-$).

Compound	Ions drawn
NaF	$[Na]^+$ and $[F]^-$ (8 electrons: 7 •, 1 ×)
MgO	$[Mg]^{2+}$ and $[O]^{2-}$ (8 electrons: 6 •, 2 ×)
$CaCl_2$	$[Ca]^{2+}$ and 2 $[Cl]^-$
Al_2O_3	2 $[Al]^{3+}$ and 3 $[O]^{2-}$
Li_3N	3 $[Li]^+$ and $[N]^{3-}$

MARK-SCHEME WORDING

- One mark per ion, each with its charge. Na^+ with 0 or 8 outer electrons; F^- with 8. Circles and brackets are not required.
- ALLOW all dots or all crosses, or dots and crosses swapped. A missing charge may lose one mark.

Often asked as: dot-and-cross of NaF, NaCl, MgO, $CaCl_2$, $BaCl_2$, aluminium oxide.

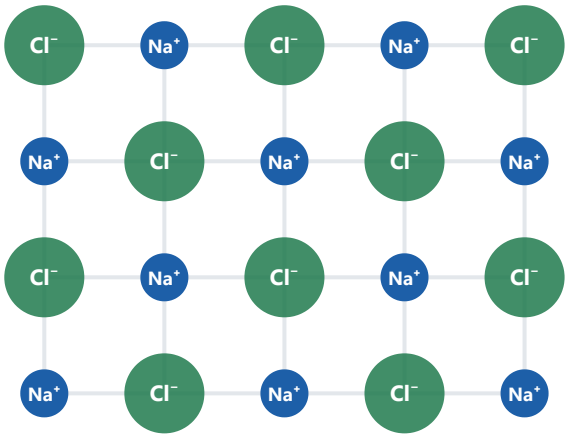
COMMON MISTAKE

- Drawing a **covalent** (shared-pair) diagram for an ionic compound, or the reverse.
- Wrong charges (Ca^+), no charges at all, or a cation with a half-filled outer shell.

SPEC 3.4 Ionic crystals are giant lattices

WHAT THIS SPEC POINT ASKS

Describe ionic crystals as giant lattices of ions.



One layer of the NaCl lattice. In 3D, each ion is surrounded by 6 ions of the opposite charge.

- An ionic solid is a **giant ionic lattice**: a **regular, repeating** 3D arrangement of a huge number of positive and negative ions, alternating so that each ion is surrounded by ions of the opposite charge.
- In NaCl each Na^+ has **6** Cl^- neighbours and each Cl^- has 6 Na^+ (a cubic arrangement). MgO has the same structure.
- The formula (NaCl) is only the **ratio** of ions; there are no molecules.
- **Brittleness**: if a force shifts one layer by one ion, ions of the **same charge** line up next to each other, repel, and the crystal splits.

MARK-SCHEME WORDING

- Drawing: "at least 9 correctly labelled ions", alternating + and −, in a regular (cubic) array.
- Say "giant ionic lattice", not "molecule" or "macromolecule".

Often asked as: draw a section of the NaCl lattice; state the structure of a solid such as barium chloride.

COMMON MISTAKE

Calling NaCl a molecule. Drawing ions of the same charge next to each other, or leaving ions unlabelled.

SPEC 3.5 Ionic bonding is a net electrostatic attraction

WHAT THIS SPEC POINT ASKS

Know that ionic bonding is the strong net electrostatic attraction between ions.

- **Ionic bonding** is the **strong electrostatic attraction between oppositely charged ions**.
- In the lattice every ion is attracted to all the oppositely charged ions around it and repelled by the like-charged ions slightly further away. The attractions are bigger than the repulsions, so there is a **net attraction**.
- The attraction acts **in all directions** (it is non-directional), and there are huge numbers of ions, so a lot of energy is needed to separate them: **high melting temperatures**.

MARK-SCHEME WORDING

- "(Strong) electrostatic attraction between **oppositely charged ions**" (or "between positive and negative ions").

Often asked as: define ionic bonding; explain why an ionic compound has a high melting temperature.

COMMON MISTAKE

Saying "attraction between atoms", or "between the nucleus and electrons" (that's covalent or metallic). Saying ionic bonds are "weak".

SPEC 3.6 How ionic charge and radius affect the strength of ionic bonding

WHAT THIS SPEC POINT ASKS

Understand how ionic radius and ionic charge affect the strength of ionic bonding.

- **Higher charge** on the ions → **stronger** attraction.
- **Smaller ions** → the centres of the ions are **closer** → **stronger** attraction.
- Stronger ionic bonding → more energy needed to separate the ions → **higher melting temperature** (and a more exothermic lattice energy, which you meet in Unit 4).

Compare	Melting temperature / °C	Reason
NaF vs NaCl	993 vs 801	F^- is **smaller** than Cl^-
NaCl vs KCl	801 vs 771	Na^+ is **smaller** than K^+
MgO vs NaCl	2852 vs 801	Mg^{2+} and O^{2-} are **doubly charged** (and Mg^{2+} is smaller)
$BaCl_2$ vs CsCl	962 vs 645	Ba^{2+} has a **bigger charge** and is **smaller** than Cs^+

WORKED EXAMPLE

Worked example: a 4-mark comparison

Explain why barium chloride has a higher melting temperature than caesium chloride.

1. The barium **ion** has a greater charge (2+) than the caesium **ion** (1+).
2. The barium ion is **smaller** than the caesium ion.
3. So the **electrostatic attraction** between the ions is **stronger** in $BaCl_2$.
4. **More energy** is needed to overcome it, so the melting temperature is higher.

MARK-SCHEME WORDING

- Name the **ions** and compare **charge** and **size**, then "stronger electrostatic attraction between the ions", then "more energy needed".
- Comparing MgO with $MgSO_4$: "oxide ion is smaller than the sulfate ion" so stronger attraction between Mg^{2+} and O^{2-}. Comparisons with sulfur or sulfide are not awarded.
- "Higher charge density" is accepted.

Often asked as: compare the melting temperatures of two ionic compounds; MCQ ranking lattice strength.

COMMON MISTAKE

- Writing about **atoms** ("magnesium has a higher charge"): it's the magnesium **ion**.
- Using atomic radius, electronegativity or intermolecular forces when the question says "consider the ions".
- Confusing ionic charge with nuclear charge, or ionisation energy with bond strength.

SPEC 3.7 Trends in ionic radius

WHAT THIS SPEC POINT ASKS

Understand the trends in ionic radius down a group, and across an isoelectronic series such as N^{3-} to Al^{3+}.

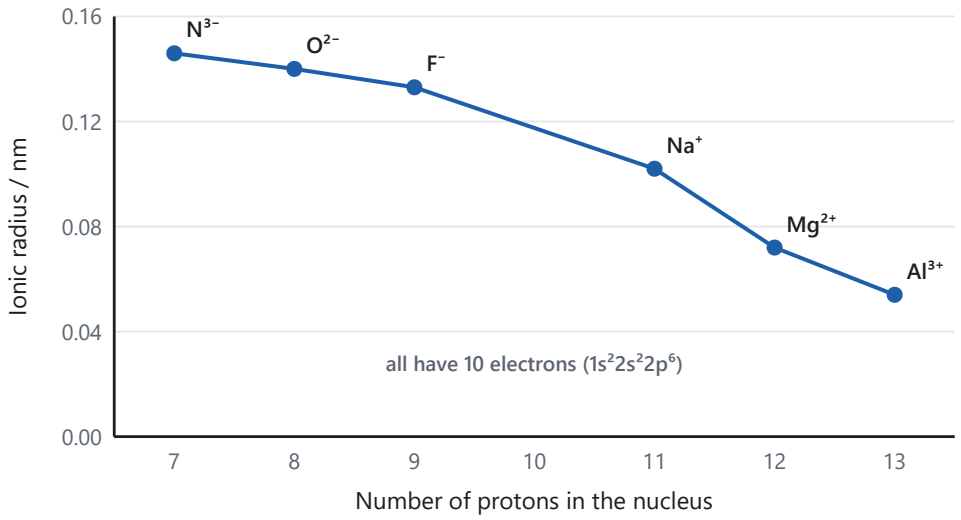
Ions compared with their atoms

- A **cation is smaller** than its atom: it has lost its whole outer shell, and the same number of protons now pulls on fewer electrons. Na atom 0.186 nm → Na^+ 0.102 nm.
- An **anion is bigger** than its atom: extra electrons increase the repulsion between electrons, and the same nuclear charge holds more electrons. Cl atom 0.099 nm → Cl^- 0.181 nm.

Down a group

Ionic radius **increases** down a group (Li^+ 0.076 < Na^+ 0.102 < K^+ 0.138 nm; F^- 0.133 < Cl^- 0.181 < Br^- 0.196 < I^- 0.220 nm), because each ion has **one more shell** of electrons.

An isoelectronic series



Isoelectronic ions N^{3-} to Al^{3+} (approximate radii; values differ slightly between data sources).

- N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+} and Al^{3+} are **isoelectronic**: all have **10 electrons**, $1s^2\ 2s^2\ 2p^6$, with the same shielding.
- From N^{3-} to Al^{3+} the **number of protons increases** from 7 to 13. The bigger nuclear charge pulls the same 10 electrons **closer**, so the **radius decreases**.

MARK-SCHEME WORDING

- Isoelectronic trend: "ionic radius **decreases** as atomic number increases" (a **trend**, not a comparison of two ions), "same number of electrons", "more protons" so electrons are pulled closer. DO NOT AWARD "atomic radius".
- Na^+ smaller than F^-: same number of electrons, but Na^+ has **more protons**, so the electrons are attracted more strongly. ALLOW "greater nuclear charge". IGNORE shielding.

Often asked as: plot the radii of N^{3-} to Al^{3+} and read off a value; describe and explain the trend; explain why Na^+ is smaller than F^-.

COMMON MISTAKE

- Comparing just N^{3-} and Al^{3+} when a **trend** for the whole series is asked.
- Writing "atomic radius" or "smaller atom" for an ion.
- Saying shielding changes across an isoelectronic series: it doesn't; only the number of protons changes.

⚠ Beyond the spec · not part of the Edexcel IAL specification

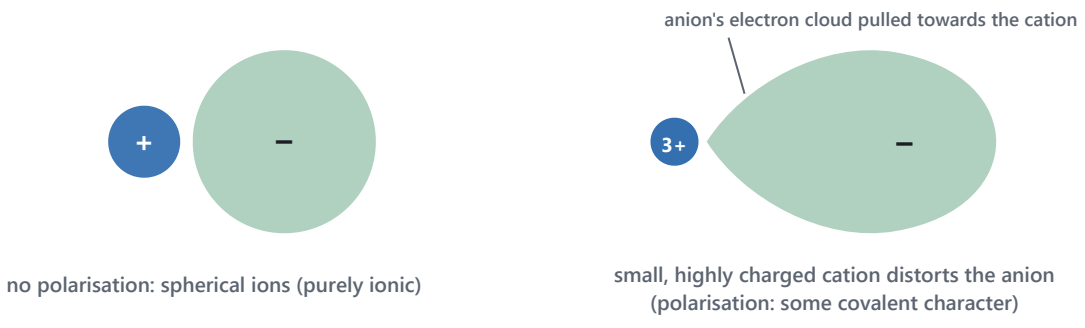
Examined in past papers even though the spec doesn't list it.

Reading an unknown ionic radius from a line of best fit through the series (a range is accepted) (WCH11 Jan 2023).

SPEC 3.8 Polarisation of ions

WHAT THIS SPEC POINT ASKS

Understand what "polarisation" means for ions.



Left: the purely ionic model. Right: a polarised anion.

- The simple ionic model treats ions as **perfect spheres** of charge.
- In reality a **cation attracts the electron cloud** of a nearby anion and **distorts** it towards itself. This is **polarisation**.
- Some electron density now sits **between** the two ions, a bit like a shared pair. So the bond gets some **covalent character**.
- The more the anion is polarised, the more covalent the compound: lower melting temperature than expected, or even a molecular structure ($AlCl_3$ sublimes at about 180 °C; AlF_3 stays solid until about 1290 °C).
- In Unit 4 you'll see the evidence: lattice energies calculated from the pure ionic model don't match experimental values for polarised compounds.

MARK-SCHEME WORDING

- "The cation **distorts / polarises** the electron cloud of the anion", giving "covalent character".

Often asked as: explain why aluminium chloride has covalent character; why fluoride is hard to polarise (3.9).

COMMON MISTAKE

Saying the cation is polarised. In this unit it is the **anion's** electron cloud that is distorted, by the cation.

SPEC 3.9 Polarising power and polarisability

WHAT THIS SPEC POINT ASKS

Understand that the polarising power of a cation depends on its radius and charge, and that the polarisability of an anion also depends on its radius and charge.

	Bigger effect when the ion is…	Examples (increasing)
Polarising power of a cation	**small** and **highly charged** (high charge density)	$K^+ < Na^+ < Li^+$; $Na^+ < Mg^{2+} < Al^{3+}$; $Ba^{2+} < Ca^{2+} < Be^{2+}$
Polarisability of an anion	**large** and **highly charged**	$F^- < Cl^- < Br^- < I^-$; $Cl^- < S^{2-}$

- A small, highly charged cation concentrates its charge in a small volume, so it pulls hard on nearby electrons.
- In a large anion the outer electrons are far from its nucleus and well shielded, so they are easily pulled away. Extra negative charge adds to this.
- **Most covalent character**: small, highly charged cation + large, highly charged anion (e.g. AlI_3, $BeCl_2$, LiI). **Most ionic**: large, singly charged cation + small, singly charged anion (e.g. CsF, KF).

WORKED EXAMPLE

Worked example: ranking covalent character

Put LiF, LiI and KI in order of increasing covalent character.

Li^+ is smaller than K^+, so it is more polarising. I^- is larger than F^-, so it is more polarisable.

KI (large cation) < LiF (small anion) < **LiI** (small cation **and** large anion) has the most covalent character.

(KI vs LiF can be argued either way; questions usually avoid that pair or give data.)

MARK-SCHEME WORDING

- Covalent character in $AlCl_3$: "the aluminium ion is (very) **small** and **highly charged**", so it "**polarises / distorts** the chloride ion / anion".
- Why F^- isn't polarised: "the fluoride ion has a **single charge** and a **small** ionic radius", so "the electron cloud is difficult to distort". The reverse argument for large, highly charged anions is allowed.
- $BeCl_2$ vs $CaCl_2$: Be^{2+} is smaller / has a higher charge density, so it is more polarising (or: Be is more electronegative).

Often asked as: explain covalent character in $AlCl_3$ or $BeCl_2$; why the fluoride ion isn't easily polarised; MCQ on which cation is most polarising.

COMMON MISTAKE

- Using the **cation's** size to explain the **anion's** polarisability.
- Saying fluoride is "large with a small charge". Not using the word **distort**.

SPEC 3.10 Covalent bonding and the evidence for it

WHAT THIS SPEC POINT ASKS

Understand that a covalent bond is the strong electrostatic attraction between two nuclei and the shared pair of electrons between them, based on evidence from (i) the physical properties of giant atomic structures and (ii) electron density maps of simple molecules.

What a covalent bond is

- A **covalent bond** is the **strong electrostatic attraction between the two nuclei and the shared pair of electrons between them**.

- Each atom usually contributes one electron to the pair. The shared pair sits between the nuclei and is attracted to both, which holds them together.
- A double bond is two shared pairs; a triple bond is three.

(i) Evidence from giant atomic (giant covalent) structures

- **Diamond**, **graphite**, **silicon** and **silicon dioxide** (SiO_2) are giant lattices of atoms joined by covalent bonds.
- They have **very high melting temperatures**, are **hard** (except graphite), **insoluble**, and (except graphite) **don't conduct**.
- To melt them you must break **many strong covalent bonds**, which needs a lot of energy. So covalent bonds must be **strong**. Their fixed shapes (e.g. tetrahedral around each atom in diamond and SiO_2) show the bonds are **directional**.

(ii) Evidence from electron density maps

- In an electron density map of H_2 or Cl_2, the contours **surround both nuclei** together, and the electron density stays **high in the region between the nuclei**.
- That build-up of electrons between the nuclei is the shared pair. Compare NaCl (3.1), where density falls to almost zero between the ions.
- In HCl the contours are pulled towards the Cl atom, showing a **polar** bond (3.14).

MARK-SCHEME WORDING

- Definition: "(electrostatic) **attraction**" between "the shared pair of electrons" and "the **nuclei** (of the two atoms)". Both the word *attraction* and the nuclei are needed.
- Electron density map of a covalent molecule: contours drawn around **both** nuclei, enclosed by a shared contour; not two separate circles.
- Giant covalent: "giant lattice", "many strong covalent bonds", "a lot of energy to break them".

Often asked as: define a covalent bond (often poorly answered); sketch or interpret an electron density map; explain the high melting temperature of SiO_2 or carbon.

COMMON MISTAKE

- Defining a covalent bond as "a shared pair of electrons" with no attraction and no nuclei.
- Mentioning ions, metallic bonding or intermolecular forces for a giant covalent structure.
- Drawing a contour map that looks like two separate atoms.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

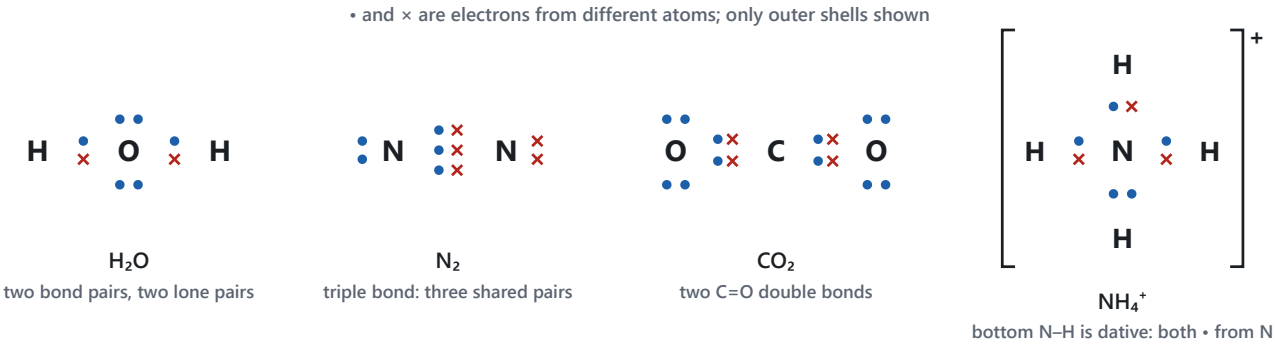
- **σ and π bonds** (spec 5.1): a σ bond is the head-on overlap of orbitals between the nuclei; a π bond is the sideways overlap of p orbitals, above and below the σ bond. A double bond is one σ + one π (WCH11 Jan 2022).
- SiO_2 is giant covalent: each Si bonded to 4 O atoms tetrahedrally, each O to 2 Si; no mobile electrons, so it doesn't conduct (WCH11 Oct 2021).

SPEC 3.11 Dot-and-cross diagrams for covalent substances, including dative bonds

WHAT THIS SPEC POINT ASKS

Draw dot-and-cross diagrams for covalent substances: (i) molecules with single, double and triple bonds; (ii) species with dative covalent (coordinate) bonds, including Al_2Cl_6 and the ammonium ion.

(i) Single, double and triple bonds



Draw every outer electron. Each shared pair has one • and one × (except a dative pair). Lone pairs belong to one atom.

Rules:

1. Use **dots** for one atom's electrons and **crosses** for the other's.
2. Put each **shared pair** between the two atoms (in the overlap if you draw circles). One pair = single bond, two = double, three = triple.
3. Show **all the non-bonding (lone) pairs**, so you can count 8 outer electrons round each atom (2 for H).
4. Then count: every atom's own electrons must add up to its group number.

Molecule	Bonds	Lone pairs
Cl_2	one single bond	3 on each Cl
H_2O	two O–H	2 on O

Molecule	Bonds	Lone pairs
NH_3	three N–H	1 on N
CH_4	four C–H	none
O_2	**O=O** (two shared pairs)	2 on each O
CO_2	**two C=O**	2 on each O, none on C
C_2H_4	C=C and four C–H	none
N_2	**N≡N** (three shared pairs)	1 on each N

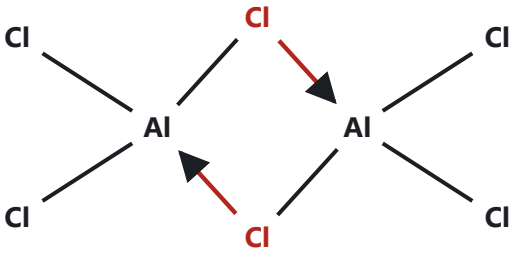
Exceptions to the octet rule

- **Electron-deficient** central atoms have fewer than 8: $BeCl_2$ (4 electrons round Be), BCl_3 and $AlCl_3$ (6 round B or Al).
- **Expanded octets**: elements in Period 3 and below can have more than 8, because they have d orbitals available: PCl_5 (10 round P), SF_6 (12 round S). Nitrogen can't do this, so NCl_5 doesn't exist.

(ii) Dative covalent (coordinate) bonds

- A **dative covalent bond** is a shared pair of electrons in which **both electrons come from the same atom**.
- You need a **donor** with a **lone pair** and an **acceptor** with an **empty orbital** (often an electron-deficient atom, or H^+).
- Show it with an **arrow from the donor atom to the acceptor atom**. In a dot-and-cross diagram, both electrons in that pair are the same symbol (both from the donor).
- Once formed, a dative bond is **identical** to any other covalent bond: all four N–H bonds in NH_4^+ are the same.

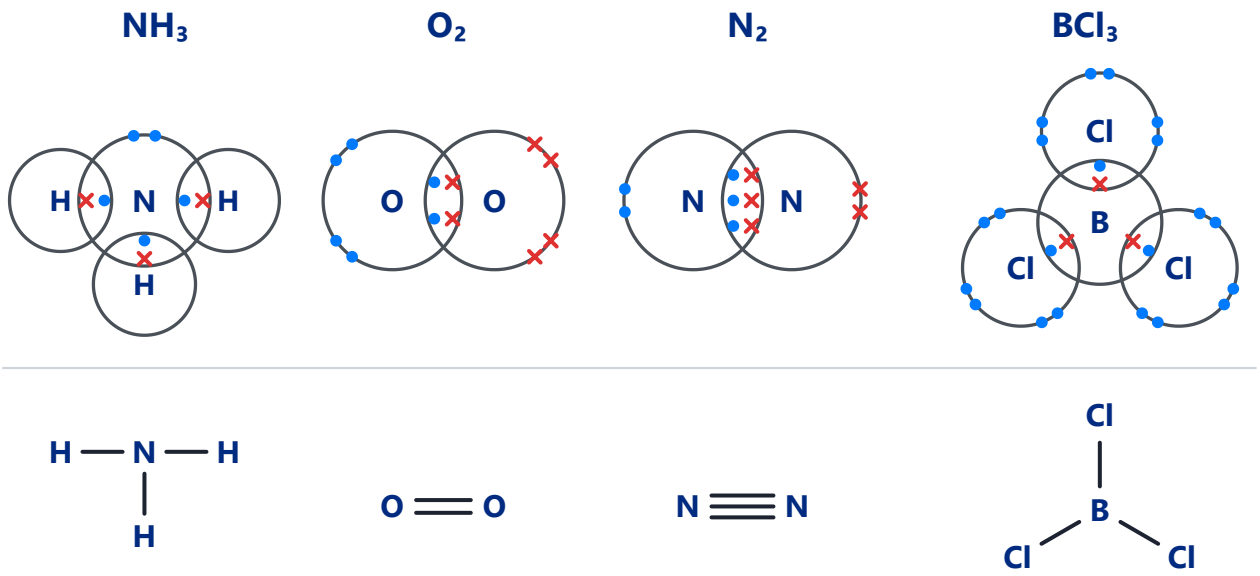
Species	Donor → acceptor
NH_4^+	lone pair on N of NH_3 → H^+
H_3O^+	lone pair on O of H_2O → H^+
Al_2Cl_6	lone pair on a Cl of one $AlCl_3$ → the Al of the other $AlCl_3$ (two dative bonds)
NH_3BF_3	lone pair on N → B (which had only 6 electrons)



arrows: dative bonds, a lone pair on Cl donated to Al

Al_2Cl_6: two $AlCl_3$ units joined by two bridging Cl atoms. Each Al ends up with 8 outer electrons. There is no Al–Al bond.

- $AlCl_3$ has only 6 electrons round Al. At lower temperatures in the vapour, two $AlCl_3$ molecules join: a lone pair on a Cl of each molecule is donated to the Al of the other, giving the **dimer Al_2Cl_6**.



Dot-and-cross diagrams of NH_3, O_2, N_2 and BCl_3 with their displayed formulae. Ammonia has three bonding pairs and one lone pair; oxygen shares two pairs and nitrogen three; boron in BCl_3 has only six outer electrons.

MARK-SCHEME WORDING

- Dot-and-cross: any combination of dots, crosses or circles; inner shells ignored. $BeCl_2$: "2 single covalent bonds" and "6 non-bonding electrons on each chlorine"; no lone pair on Be. An ionic diagram for a

covalent molecule scores 0.

- Dative bond definition: "a (shared) pair of electrons" where "both electrons come from the same atom".
- Dative arrow goes **from the donor (Cl) to the acceptor (Al)**. In a dot-and-cross, the two dative electrons "must appear the same".
- NH_4^+: square brackets and a + charge, with one N–H shown as coordinate.
- Expanded octet (ClF_3): "10 electrons / 5 pairs in the outer shell of chlorine", "has expanded its octet".

Often asked as: dot-and-cross of N_2, Br_2, CCl_4, SCl_2, $BeCl_2$, NCl_3, $B(OH)_3$, ClF_3, $SbCl_5$, SF_4; Al_2Cl_6 with dative bonds; NH_4^+; define a dative bond; why N can't form NCl_5.

COMMON MISTAKE

- Lone pairs left off Cl or Br; an extra lone pair added to Be or B.
- A single bond in N_2, or drawing a halogen molecule when N_2 was asked.
- Dative arrows from Al to Cl; an Al–Al bond; extra Cl^- ions in Al_2Cl_6.
- Drawing NH_3 when NH_4^+ was asked, or leaving off the charge. Calling the dative pair a "lone pair" after it has bonded.
- Merging O and H into one symbol in $B(OH)_3$; diagrams too small to count.

⚠ Beyond the spec · not part of the Edexcel IAL specification

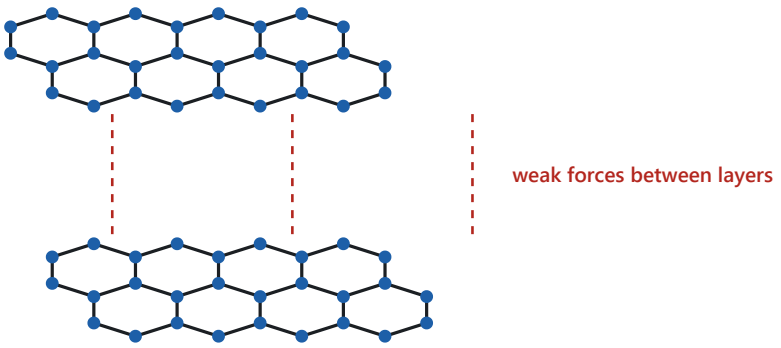
Examined in past papers even though the spec doesn't list it.

- Expanded octets in unfamiliar molecules: ClF_3, $SbCl_5$, SF_4, S_2F_{10} (WCH11 Oct 2019, Jan 2021, Jan 2025).
- Why nitrogen can't form NCl_5: it has **no 2d orbitals** (no available orbitals in the second shell), so it can't expand its octet (WCH11 Jan 2021).
- Negative ions: add the extra electron(s) to the central atom's count, e.g. GeH_3^- has a lone pair on Ge (WCH11 Oct 2022). Oxoanions such as SO_4^{2-} (WCH11 Oct 2025).

SPEC 3.12 Giant lattices of carbon: graphite, diamond and graphene

WHAT THIS SPEC POINT ASKS

Describe the giant structures of carbon (graphite, diamond and graphene) and discuss their uses.

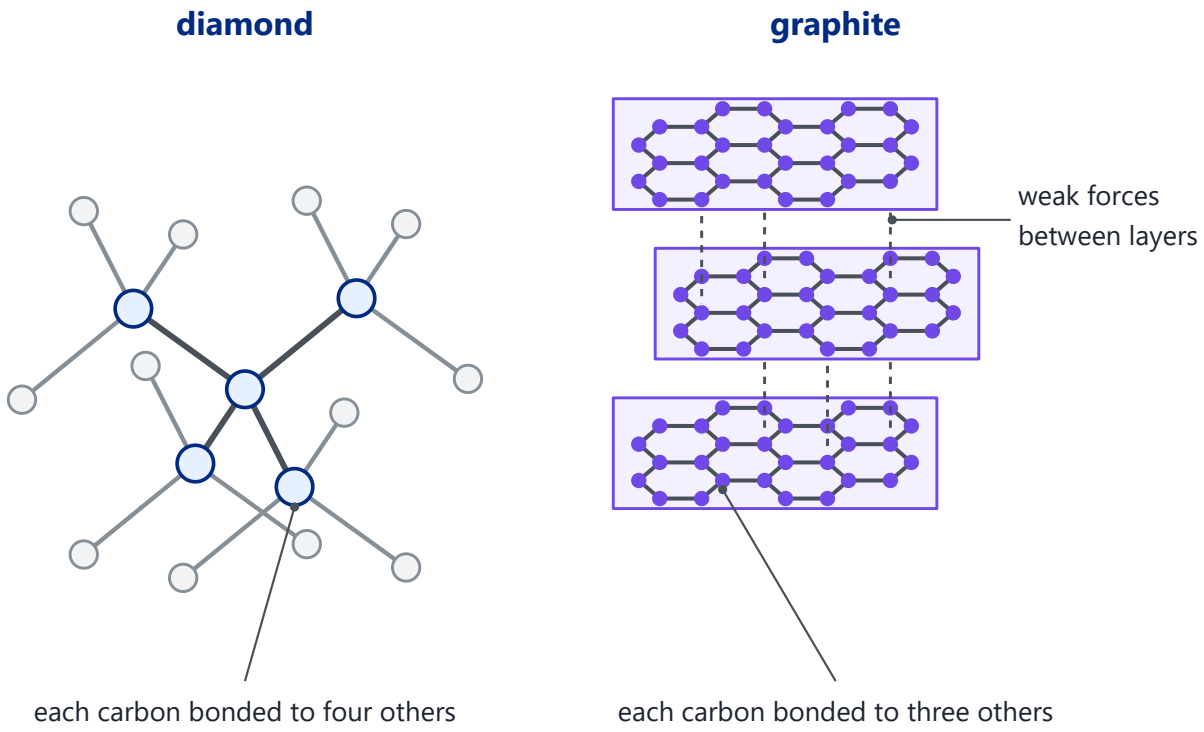


Graphite: layers of hexagons. When drawing it, show at least two layers with at least three hexagons each, and label the forces between the layers.

	Diamond	Graphite	Graphene
Structure	giant covalent; each C bonded to **4** others, **tetrahedral** (109.5°)	giant covalent **layers** of hexagons; each C bonded to **3** others (120°)	a **single layer** of graphite, one atom thick
Delocalised electrons	**none**: all 4 outer electrons in bonds	**one per C**, free to move along the layers	one per C, across the sheet
Between layers	n/a	**weak London forces**; layers 0.335 nm apart	n/a
Electrical conductivity	**doesn't** conduct	**conducts** (along the layers)	conducts very well
Hardness	**very hard**	**soft**, slippery: layers slide	very strong for its mass; flexible
Density	3.5 g cm^{-3} (higher)	2.3 g cm^{-3} (lower: layers far apart)	very light; almost transparent
Uses	cutting tools, drill tips, jewellery	electrodes, lubricant, pencils	electronics, touchscreens, light strong composites (e.g. sports equipment), conductive inks, sensors

- **Both** diamond and graphite have **very high melting temperatures**: graphite's layers are held by weak forces, but to **melt** it you must break the **strong covalent bonds within the layers**.

- Each use should be linked to a property: graphite **electrodes** because it conducts, has a high melting temperature and is fairly unreactive; **lubricant** because the layers slide; diamond **cutting tools** because it is the hardest substance.



Diamond (left): each carbon is bonded to four others in a rigid tetrahedral network. Graphite (right): each carbon is bonded to three others in flat hexagonal layers, which are stacked with weak forces between them.

MARK-SCHEME WORDING

- Diamond: "each carbon covalently bonded to **four** others"; "no delocalised electrons", so no conduction; hard. Graphite: "layers", "each carbon bonded to **three** others", "one delocalised electron per atom", so it conducts; "weak forces between layers", so they slide.
- Differences must be **properties** (conductivity, hardness, density). **Melting temperature is not accepted** as a difference between diamond and graphite.
- Graphene: "a single layer of graphite"; a use **with** the property that suits it (strong, light, conducts, transparent). "Lubricant" alone is not credited for graphene.
- Drawing graphite: "at least two layers of hexagons (at least 3 hexagons per layer)" and "labelled forces of attraction between the layers".

Often asked as: compare diamond and graphite (properties and structure); relate graphene to graphite and give a use; why graphite is used as an electrode; draw graphite layers; explain the high melting temperature of carbon.

COMMON MISTAKE

- Calling graphite or diamond "molecular", or saying diamond has "intermolecular forces".
- Mixing up the **weak forces between layers** with **weak bonds within** the layers.
- Writing about diamond, then graphite, in separate paragraphs in a "compare" question: compare in the same sentence.
- Confusing graphene with graphite.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Showing the electrons in diamond as a dot-and-cross section: each C with four shared pairs (WCH11 Jun 2024).
- Graphite as the electrode in aluminium extraction at about 950 °C: giant covalent, high melting temperature, delocalised electrons carry the current (WCH11 Jun 2023).

SPEC 3.13 Electronegativity

WHAT THIS SPEC POINT ASKS

Understand what electronegativity means for atoms in a covalent bond.

- **Electronegativity** is the **ability of an atom to attract the bonding pair of electrons** in a covalent bond.
- It is measured on the **Pauling scale**. Fluorine is the most electronegative element (4.0). Noble gases have no value (they don't usually bond).

	Group 1	2	13	14	15	16	17
Period 1	H 2.1						
Period 2	Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Period 3	Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
Period 4	K 0.8	Ca 1.0					Br 2.8

Trends

- **Increases across a period**: more protons (bigger nuclear charge), similar shielding, smaller atom, so the bonding pair is attracted more strongly.

- **Decreases down a group**: more shells, more shielding and a larger distance from the nucleus to the bonding pair, so the attraction is weaker.

MARK-SCHEME WORDING

- "The ability of an atom to attract the **bonding pair** of electrons (in a covalent bond)." "Attract electrons" alone is too vague.

Often asked as: define electronegativity; use values to decide which end of a bond is δ−.

COMMON MISTAKE

Defining electronegativity as "how easily an atom gains an electron" (that's electron affinity) or leaving out "in a covalent bond / bonding pair".

SPEC 3.14 The bonding continuum: from covalent to ionic

WHAT THIS SPEC POINT ASKS

Know that ionic and covalent bonding are the two extremes of a continuum, and explain this through electronegativity differences: polar bonds, polar molecules, and ionic bonding when the difference is large enough.

Electronegativity difference	Bond	Example
zero	**pure (non-polar) covalent**: pair shared equally	H–H, Cl–Cl
small	**polar covalent**: pair pulled towards the more electronegative atom, which gets **δ−**; the other gets **δ+**	$H^{\delta+}$–$Cl^{\delta-}$ (difference 0.9)
large	**ionic**: the electron is effectively transferred	NaCl (2.1), NaF (3.1)

- Write polar bonds as **$H^{\delta+}$–$Cl^{\delta-}$**. The δ− goes on the **more electronegative** atom.
- There is **no sharp cut-off** between the types. A polar covalent bond has some ionic character; an ionic bond with a polarised anion has some covalent character (3.8). Bonds sit somewhere on a **continuum**.
- The bigger the difference, the more **ionic character**: $BaCl_2$ (3.0 − 0.9 = 2.1) is more ionic than $BeCl_2$ (3.0 − 1.5 = 1.5).

MARK-SCHEME WORDING

- "Ionic and covalent are extremes of a **continuum**"; the electronegativity difference decides where a bond lies. Both compounds have "some ionic and some covalent character"; one is "more ionic / less covalent".
- State which element is **more electronegative**, then put δ− on it.

Often asked as: use electronegativity values to compare two compounds; mark δ+ and δ− on a bond.

COMMON MISTAKE

- Treating a difference as a fixed cut-off ("1.7 or more is ionic"). Examiners mark that as a misconception.
- Putting δ+ and δ− the wrong way round; drawing a dipole arrow when δ+/δ− was asked.

SPEC 3.15 Polar bonds and polar molecules

WHAT THIS SPEC POINT ASKS

Distinguish between polar bonds and polar molecules, and predict whether a molecule is likely to be polar.

- A molecule can contain **polar bonds** and still be **non-polar overall**. It is a **polar molecule** only if the bond dipoles **don't cancel**: one end of the molecule is slightly positive and the other slightly negative (a permanent dipole).
- **Symmetrical** shapes with identical outer atoms and no lone pair on the central atom cancel out.

Molecule	Polar bonds?	Shape	Polar molecule?
CO_2	yes (C=O)	linear	**no**: dipoles cancel
CCl_4, CH_4	C–Cl yes; C–H nearly non-polar	tetrahedral	**no**
BF_3	yes	trigonal planar	**no**
SF_6	yes	octahedral	**no**
H_2O	yes	bent	**yes**
NH_3	yes	trigonal pyramidal	**yes**
HCl	yes	linear (two atoms)	**yes**

Molecule	Polar bonds?	Shape	Polar molecule?
$CHCl_3$, CH_3Cl	yes	tetrahedral but not symmetrical	**yes**

Method

1. Work out the **shape** (3.18 and 3.19).
2. Find the **polar bonds** (electronegativity difference).
3. Ask whether the bond dipoles **cancel** because of symmetry.

The charged rod test

Run a thin jet of liquid from a burette past a **charged rod** (rubbed polythene or acetate). Jets of **water** and **ethanol** bend towards the rod: their molecules are polar and turn so their oppositely charged end faces the rod. A jet of **cyclohexane** isn't deflected: its molecules are non-polar. The polar jet bends towards the rod whether the rod is positive or negative.

MARK-SCHEME WORDING

- CCl_4: "chlorine is **more electronegative** than carbon (so the bonds are polar)" **but** "the molecule is not polar because the **dipoles cancel** / the molecule is **symmetrical**". IGNORE "the polar bonds cancel".
- Polar molecule: "one end slightly positive and the other slightly negative".
- CO_2: oxygen is more electronegative, so δ+ on C and δ− on each O; but linear, so non-polar.

Often asked as: is CCl_4 / CO_2 polar, and why; define a polar molecule; MCQ from jet deflection.

COMMON MISTAKE

- Arguing that a difference is "not big enough" to be polar.
- Getting the shape wrong in one part and carrying it into the polarity answer.
- Calling C–H bonds strongly polar.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Predicting polarity of an unfamiliar molecule from its shape: SF_4 (see-saw) is polar, SF_6 (octahedral) is not (WCH11 Jan 2025).
- **Induced dipoles**: the π electrons of an alkene repel the electrons in a Cl_2 or Br_2 molecule, distorting its electron cloud and inducing a dipole (WCH11 Jan 2025; see Topic 5).

SPEC 3.16 Electron-pair repulsion theory

WHAT THIS SPEC POINT ASKS

Understand the principles of electron-pair repulsion theory, used to explain and predict the shapes of simple molecules and ions.

The principles

1. The **electron pairs** in the outer shell of the central atom (both **bonding pairs** and **lone pairs**) **repel** each other.
2. They arrange themselves **as far apart as possible**, to **minimise repulsion**. That arrangement fixes the shape.
3. **Lone pairs repel more** than bonding pairs, because a lone pair is held closer to the central nucleus and spreads out more. The order is: **lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair**.
4. Each lone pair squeezes the bonds closer together, cutting the bond angle by **roughly 2.5°** (109.5° → 107° → 104.5°).
5. A **double (or triple) bond** counts as **one** bonding region for working out the shape.
6. The **shape is named from the atoms** only, but the lone pairs still affect it.

Counting the pairs

For a central atom bonded only by single bonds:

- electrons = central atom's outer electrons + **1 for each bonded atom** + **1 for each negative charge** − **1 for each positive charge**
- pairs = electrons ÷ 2; **lone pairs** = pairs − bonding pairs.
- If double bonds are present, draw the dot-and-cross diagram first and count each double bond as one region.

Total pairs (regions)	Arrangement	Angle with no lone pairs
2	linear	180°
3	trigonal planar	120°
4	tetrahedral	109.5°
5	trigonal bipyramidal	120° and 90°
6	octahedral	90°

Explaining the shape of ammonia

1. Nitrogen has five outer electrons. Three are shared with the three hydrogen atoms, which leaves one lone pair. So there are **3 bonding pairs and 1 lone pair**.
2. The four electron pairs repel each other and move as far apart as possible, into a tetrahedral arrangement.
3. The lone pair repels more strongly than the bonding pairs, so it pushes the N–H bonds slightly closer together.
4. The shape is **trigonal pyramidal** and the H–N–H bond angle is **107°**, smaller than the 109.5° of a regular tetrahedron.

Exam answers need all of these: the number of each kind of pair, that pairs repel to get as far apart as possible, that lone pairs repel more (or, with no lone pairs, that the pairs repel equally), and then the name of the shape and the angle.

MARK-SCHEME WORDING

- "Electron pairs repel (each other)" and "get as far apart as possible / to minimise repulsion". ALLOW "maximum separation".
- "Lone pairs repel more than bonding pairs."
- Teach and use **electron pairs**. One recent mark scheme ignored "bonds" or "electron regions"; another accepted regions of electron density for a double bond. So say "pairs of electrons", and say a double bond "counts as one region".

Often asked as: inside every shapes question (3.18, 3.19).

COMMON MISTAKE

- "The **atoms** (or bonds) repel", or "the molecule moves to minimise repulsion". It is the **electron pairs** that repel.
- Saying lone pairs repel "the atoms".
- Quoting "2.5° per lone pair" without the reason (lone pairs repel more).

SPEC 3.17 Bond length and bond angle

WHAT THIS SPEC POINT ASKS

Understand the terms bond length and bond angle.

- **Bond length**: the distance between the **nuclei** of two atoms joined by a covalent bond. It is where attraction (nuclei to the shared pair) and repulsion (nucleus to nucleus) balance.
- **Bond angle**: the angle between **two bonds** that meet at the **same atom** (between the lines joining the nuclei).
- More shared pairs pull the nuclei closer: multiple bonds are **shorter and stronger**.

Bond	Length / nm	Mean bond enthalpy / kJ mol^{-1}
C–C	0.154	347
C=C	0.134	612
C≡C	0.120	838

MARK-SCHEME WORDING

- Bond length: "distance between the **nuclei** (of two bonded atoms)" (not "between the atoms").
- Bond angle: label it at the **central** atom, between two bonds.

Often asked as: definitions; MCQ comparing bond lengths; label an angle on a diagram.

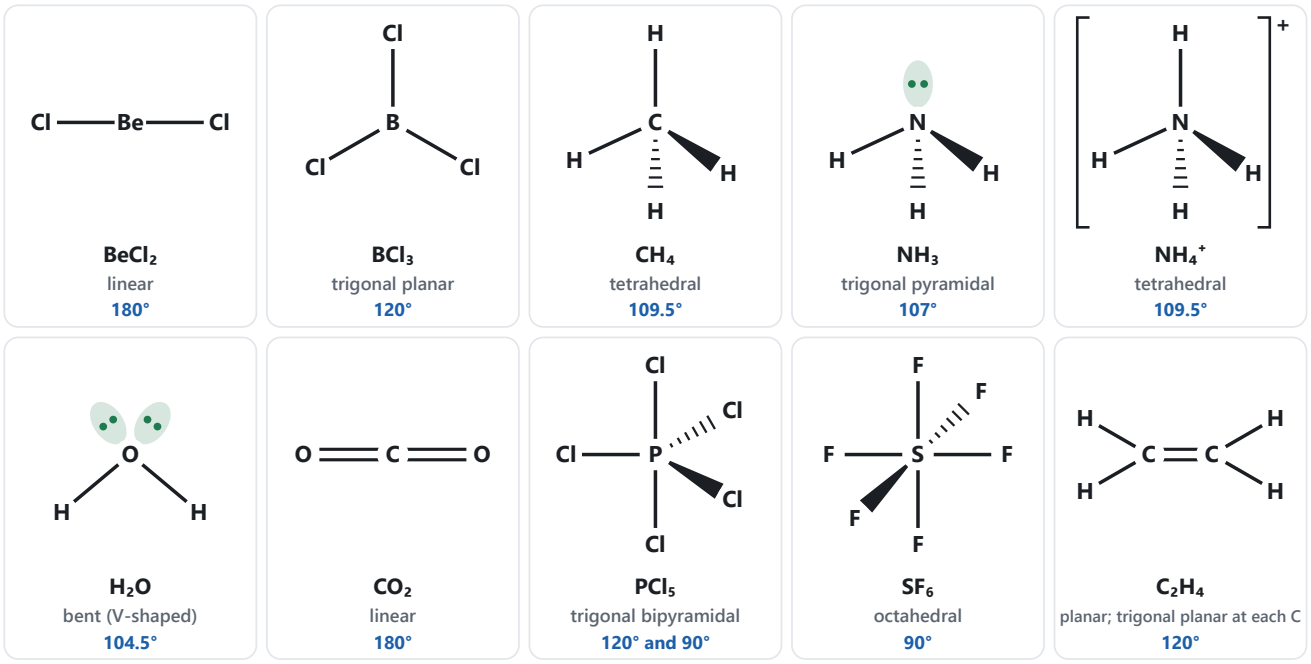
COMMON MISTAKE

Labelling a bond angle in the wrong place, or giving two different angles for one label (the second one cancels the mark).

SPEC 3.18 Shapes of named molecules and ions

WHAT THIS SPEC POINT ASKS

Know and explain the shapes and bond angles of $BeCl_2$, BCl_3, CH_4, NH_3, NH_4^+, H_2O, CO_2, gaseous PCl_5, SF_6 and C_2H_4.



A solid wedge comes towards you; a hashed wedge goes away; plain lines are in the plane of the page. Green lobes are lone pairs.

Species	Bonding pairs	Lone pairs	Shape	Bond angle
$BeCl_2$	2	0	**linear**	180°
BCl_3	3	0	**trigonal planar**	120°
CH_4	4	0	**tetrahedral**	109.5°
NH_4^+	4 (one dative)	0	**tetrahedral**	109.5°
NH_3	3	1	**trigonal pyramidal**	107°
H_2O	2	2	**bent / V-shaped**	104.5°
CO_2	2 double bonds (2 regions)	0	**linear**	180°
PCl_5 (gas)	5	0	**trigonal bipyramidal**	120° (equatorial) and 90° (axial to equatorial)
SF_6	6	0	**octahedral**	90°
C_2H_4	3 regions round each C (2 C–H, 1 C=C)	0	**trigonal planar** at each C; the whole molecule is flat	120°

WORKED EXAMPLE

Worked example: the full explanation for NH_3

1. N has 5 outer electrons; 3 H atoms add 3 → 8 electrons = **4 pairs**: **3 bonding pairs and 1 lone pair**.
2. The electron pairs **repel** and get **as far apart as possible**: a tetrahedral arrangement of pairs.
3. The **lone pair repels more** than the bonding pairs, so it squeezes the N–H bonds together.
4. Shape: **trigonal pyramidal**; bond angle **107°** (less than 109.5°).
   When NH_3 gains H^+ to become NH_4^+, the lone pair becomes a bonding pair: **4 bonding pairs**, all repel equally, **tetrahedral, 109.5°**.

WORKED EXAMPLE

Worked example: why CH_4, NH_3 and H_2O differ

All three have **4 electron pairs** round the central atom.

CH_4: 4 bonding pairs, 0 lone pairs → 109.5°.

NH_3: 3 bonding pairs, **1** lone pair → 107°.

H_2O: 2 bonding pairs, **2** lone pairs → 104.5°.

Lone pairs repel more than bonding pairs, so **each extra lone pair reduces the angle**.

EXAM TIP

Sketch 3D shapes with wedges and dashes, label the angle **between two bonds at the central atom**, and give **one** value per angle. Write "trigonal **bi**pyramidal" in full: "trigonal pyramidal" is a different shape.

MARK-SCHEME WORDING

- Standard 2-mark answer: the angle (e.g. 109.5°) **and** "(four) electron pairs / bonding pairs repel to get maximum separation / minimum repulsion".
- Always state the numbers of **bonding pairs and lone pairs**.
- Water-like angles: 103° to 104.5° accepted in recent mark schemes. Learn **104.5°**.
- CO_2: two C=O regions, no lone pairs on C → linear, 180°.
- PCl_5: "trigonal bipyramidal", "**90°** and **120°**". SF_6: "octahedral", "90°".

Often asked as: a table of bonding pairs, lone pairs, shape and angle (CH_4, NH_3, H_2O); explain why the angles differ; explain the shape of a named molecule; NH_3 vs NH_4^+.

COMMON MISTAKE

- Forgetting a lone pair, which turns NH_3 or NCl_3 into "trigonal planar".
- Calling the dative pair in NH_4^+ a lone pair.
- Mentioning lone pair–lone pair repulsion in a molecule with only **one** lone pair (NH_3, PCl_3).
- "Trigonal pyramidal" for BCl_3 or $AlCl_3$ (no lone pair: trigonal planar).
- Not knowing both angles in PCl_5.

SPEC 3.19 Predicting the shapes of other molecules and ions

WHAT THIS SPEC POINT ASKS

Apply electron-pair repulsion theory to predict the shapes and bond angles of molecules and ions similar to those in 3.18.

Method (show every step)

1. Find the central atom's outer electrons (its group number).
2. Add one for each atom bonded by a single bond; add the negative charge, or subtract the positive charge.
3. Halve it to get the **number of pairs**; split them into **bonding pairs** and **lone pairs**.
4. Pick the arrangement from the number of pairs (3.16), then name the shape from the atoms.
5. Adjust the angle for lone pairs (about 2.5° less per lone pair, from 109.5°).

WORKED EXAMPLE

Worked example: four analogues

PCl_4^+: 5 + 4 − 1 = 8 → 4 pairs, 4 bonding, 0 lone → **tetrahedral, 109.5°** (like NH_4^+).

PCl_6^-: 5 + 6 + 1 = 12 → 6 pairs, 6 bonding → **octahedral, 90°** (like SF_6).

NCl_3: 5 + 3 = 8 → 4 pairs, 3 bonding, 1 lone → **trigonal pyramidal, 107°** (like NH_3).

ClF_3: 7 + 3 = 10 → 5 pairs, 3 bonding, 2 lone → **T-shaped**, about 87.5° (just under 90°). The two lone pairs sit in the equatorial positions, where they have most room.

Analogues to recognise

Species	Like	Bonding pairs	Lone pairs	Shape	Angle
H_3O^+, NCl_3, PCl_3, GeH_3^-	NH_3	3	1	trigonal pyramidal	107°
NH_2^-, SCl_2, HOCl (at O)	H_2O	2	2	bent	104.5°
BF_4^-, $SiCl_4$, CCl_4, PCl_4^+	CH_4	4	0	tetrahedral	109.5°
BF_3, $AlCl_3$	BCl_3	3	0	trigonal planar	120°
CS_2	CO_2	2 regions	0	linear	180°
CO_3^{2-}, NO_3^-	BCl_3	3 regions	0	trigonal planar	120°
SO_4^{2-}	CH_4	4 regions	0	tetrahedral	109.5°
PF_5, $SbCl_5$	PCl_5	5	0	trigonal bipyramidal	120° and 90°
PCl_6^-, S_2F_{10} (at each S)	SF_6	6	0	octahedral	90°
SF_4, $TlCl_4^{3-}$	(new)	4	1	see-saw	about 102° and 173° (below 120° and 180°)
ClF_3	(new)	3	2	T-shaped	about 87.5°
XeF_4	(new)	4	2	square planar	90°

Predicting the shape of an ion: H_3O^+

1. Start with the central atom's outer electrons: oxygen has 6.
2. Add one electron for each atom bonded by a single bond (3 H), then adjust for the charge: a 1+ charge means one electron fewer. $6 + 3 - 1 = 8$ electrons, which is **4 pairs**.
3. Three pairs bond to H atoms, so there are 3 bonding pairs and 1 lone pair: the same as ammonia.
4. H_3O^+ is trigonal pyramidal, with bond angles of about 107°.

Predicting the shape of XeF_4

1. Xenon has 8 outer electrons, and the four fluorine atoms add one each: $8 + 4 = 12$ electrons, so **6 pairs**.
2. Four pairs bond to F atoms, leaving 2 lone pairs. Six pairs point to the corners of an octahedron.
3. The two lone pairs repel each other most, so they take opposite positions, 180° apart, above and below the plane.
4. The four F atoms lie in a flat square around Xe: the shape is **square planar** and the F–Xe–F bond angle is **90°**.

MARK-SCHEME WORDING

- Quote "bond pairs **and** lone pairs" for each centre: e.g. in a ring with an O atom, "four bond pairs around C (109.5°); two bond pairs and two lone pairs around O (104.5°)".
- Shape "and justification": "electron pairs arranged to minimise repulsion", with the count of pairs.
- SF_4: see-saw, angles about 90° and 180° (about 173° accepted). $TlCl_4^{3-}$: has a **lone pair** on Tl, so see-saw.
- Deduce the **charge** of an ion such as PCl_4^+ or PCl_6^- from its shape.

Often asked as: shapes of NCl_3, PCl_3, SCl_2, GeH_3^-, HOCl, $AlCl_3$, $TlCl_3$, $TlCl_4^{3-}$, SF_4, S_2F_{10}, PCl_4^+/PCl_6^-; angles at C and O in an organic ring.

COMMON MISTAKE

- Not counting the lone pairs from the dot-and-cross diagram you've just drawn.
- Forgetting to add or subtract electrons for the ion's charge.
- Giving 180° for an O–B–O angle in $B(OH)_3$ (it's trigonal planar at B, 120°).

⚠ Beyond the spec · not part of the Edexcel IAL specification

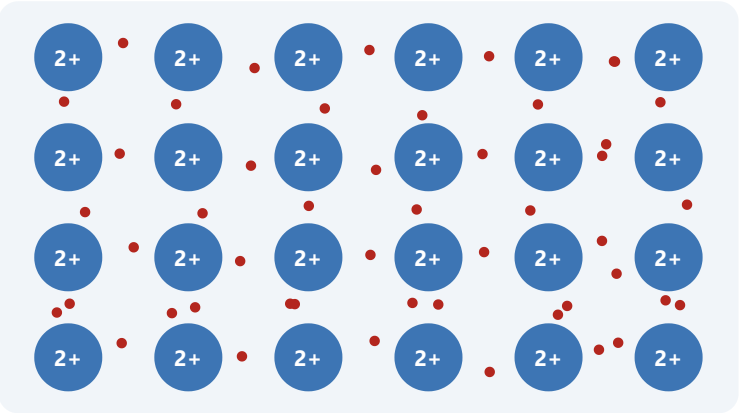
Examined in past papers even though the spec doesn't list it.

- The beryllium chloride **polymer** in the solid: each Be has four bonding pairs (two of them dative), so it is tetrahedral round Be (WCH11 Oct 2021).
- Shapes with 5 or 6 pairs including lone pairs: ClF_3, SF_4, $TlCl_4^{3-}$ (WCH11 Oct 2019, Jan 2025, Oct 2024).
- Angles in organic molecules: 109.5° at a saturated C, 104.5° at an O with two lone pairs (WCH11 Jan 2020).

SPEC 3.20 The structure of metals

WHAT THIS SPEC POINT ASKS

Understand that metals are giant lattices of metal ions in a sea of delocalised electrons.



Magnesium: a regular lattice of Mg^{2+} ions, with two delocalised electrons per ion moving freely between them.

- In a metal, each atom releases its **outer electrons**, which become **delocalised**: free to move through the whole structure, not tied to any one atom.
- What's left is a **giant lattice of positive metal ions** (cations), arranged in regular layers, held in a "**sea**" of delocalised electrons.
- The number of delocalised electrons per ion equals the ion's charge: Na gives 1 (Na^+), Mg gives 2 (Mg^{2+}), Al gives 3 (Al^{3+}).

MARK-SCHEME WORDING

- Diagram: a **regular** array of cations **with charges**, plus electrons in an **irregular** arrangement between them, about **twice as many** electrons as ions for a 2+ ion. Label the electrons or show them as e^- or dots.

Often asked as: draw a labelled diagram of calcium, silver or magnesium.

COMMON MISTAKE

Drawing too many or too few electrons, or arranging them in a regular grid so the diagram looks ionic. Calling the particles "atoms" instead of **ions**.

SPEC 3.21 Metallic bonding

WHAT THIS SPEC POINT ASKS

Know that metallic bonding is the strong electrostatic attraction between metal ions and delocalised electrons.

- **Metallic bonding** is the **strong electrostatic attraction between the positive metal ions and the delocalised electrons**.
- It acts throughout the lattice and isn't directional.
- It gets **stronger** when the ions have a **higher charge** and there are **more delocalised electrons** per ion (and when the ions are smaller). So: Na < Mg < Al.

WORKED EXAMPLE

Worked example: magnesium vs sodium

Explain why magnesium has a higher melting temperature than sodium.

1. Magnesium forms **Mg^{2+}** ions; sodium forms **Na^{+}** ions (higher charge).

2. Magnesium has **two delocalised electrons per atom**; sodium has one.

3. So the **attraction between the ions and the delocalised electrons** is **stronger** in magnesium, and more energy is needed to overcome it.

MARK-SCHEME WORDING

- "(**Strong**) electrostatic attraction between (metal) **cations / positive ions** and **delocalised electrons**". The word **strong** was required (WCH11 Jan 2023). IGNORE protons or nuclei. Any mention of another bond type scores 0.
- Mg vs Na (2 marks): **greater charge** on the ion **and more delocalised electrons**. In this mark scheme ionic size was ignored and "just a higher charge" wasn't enough.

Often asked as: describe metallic bonding in sodium or aluminium; compare the melting temperatures of two metals.

COMMON MISTAKE

Saying "free electrons" or "charge carriers" instead of **delocalised electrons**. Describing attraction between positive and **negative ions** in a metal. Leaving out "strong".

SPEC 3.22 Explaining the properties of metals

WHAT THIS SPEC POINT ASKS

Use the metallic bonding model to explain simple properties of metals, including electrical conductivity and high melting temperature.

Property	Explanation
Conduct electricity (solid **and** liquid)	the **delocalised electrons can move** through the structure; when a potential difference is applied they flow and carry the charge
High melting and boiling temperatures	the **strong attraction** between the ions and the delocalised electrons needs a lot of energy to overcome
Good **thermal** conductors	delocalised electrons carry kinetic energy quickly through the lattice
Malleable and ductile	layers of ions can **slide** over each other; the delocalised electrons move with them, so the **bonding isn't broken**

Alloys

An **alloy** is a mixture of a metal with other elements. The **ions of different sizes** disrupt the regular layers, so the layers can't slide over each other as easily: alloys are **harder and stronger** than the pure metal.

Identifying a structure from data

1. A white solid melts at 1710 °C. It does not conduct electricity as a solid or when molten, and it does not dissolve in water. What type of structure does it have?
2. The very high melting temperature rules out a simple molecular structure: something strong holds every particle in place.
3. It does not conduct when molten, so it has no ions that become mobile on melting: it is not ionic. It does not conduct as a solid, so it is not a metal.
4. It must be **giant covalent**. (These are the properties of silicon(IV) oxide.)

Check conductivity in both states: solid only tells metal and graphite apart from the rest; molten tells ionic compounds apart from giant covalent ones.

MARK-SCHEME WORDING

- Conductivity: "**delocalised electrons** that **move / flow**" in both the solid and the liquid. Any reference to **ions** moving in a metal is not awarded.
- Malleability: "layers of cations slide over each other" and "the bonding is not broken".
- Alloys: "ions of **different size**" so "layers don't slide as easily". The word **ion** is needed.
- A stated use needs a matching property (e.g. aluminium cables: good conductor, low density, ductile).

Often asked as: why a metal conducts when solid and when molten; why it is malleable; why an alloy is harder; why aluminium is used for a given job.

COMMON MISTAKE

- "Electrons are delocalised" without saying they **move**.
- Explaining a metal's conductivity by the charge on its ions alone.
- Mixing models: "delocalised electrons in the molten ionic compound" or "mobile ions in the metal".
- A general list of metal properties when the question is about one metal and one use.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Alloys: silver alloyed with copper is harder because the copper ions are a different size and stop the layers sliding (WCH11 Jun 2024).
- Choosing aluminium for a use from its conductivity, ductility and low density (WCH11 Oct 2021).

Quick checks

QUICK CHECK

What is the shape of a PCl_3 molecule, and its approximate bond angle?

- trigonal planar, 120°
- trigonal pyramidal, 107°
- tetrahedral, 109.5°
- T-shaped, 90°

Phosphorus has 5 outer electrons; three bond to Cl, leaving one lone pair. Four pairs give a tetrahedral arrangement, and the lone pair squeezes the bonds to about 107°, so the shape is pyramidal. Trigonal planar would need three pairs and no lone pair, as in BCl_3.

QUICK CHECK

Why does magnesium oxide have a much higher melting temperature than sodium chloride?

- MgO has strong covalent bonds between its atoms.
- MgO molecules have stronger intermolecular forces.
- Mg is more reactive than Na.
- Mg^{2+} and O^{2-} have higher charges and Mg^{2+} is smaller, so the ionic attraction is stronger.

Both compounds are giant ionic lattices. Doubly charged ions, closer together, attract each other much more strongly, so more energy is needed to separate them. MgO has no molecules, so "intermolecular forces" is wrong.

QUICK CHECK

Which of these molecules is polar?

- CO_2
- CCl_4
- $CHCl_3$
- BF_3

All four contain polar bonds, but in CO_2, CCl_4 and BF_3 the symmetrical shape makes the bond dipoles cancel. In $CHCl_3$ one position holds H instead of Cl, so the dipoles no longer cancel and the molecule is polar.

QUICK CHECK

Which ion has the largest radius?

- O^{2-}
- F^-
- Na^+
- Mg^{2+}

All four have 10 electrons. Oxide has the fewest protons (8), so its electrons are held least tightly and it is the biggest. Mg^{2+}, with 12 protons, is the smallest.

QUICK CHECK

Why does graphite conduct electricity but diamond does not?

- Graphite contains ions that can move between the layers.
- In graphite one electron per carbon is delocalised and can move along the layers; in diamond all outer electrons are in bonds.
- The layers in graphite slide over each other.
- Diamond has stronger covalent bonds, which trap the electrons.

Conduction needs mobile charged particles. Each carbon in graphite uses only three electrons in bonds, and the fourth is delocalised within its layer. Sliding layers explain why graphite is soft, not why it conducts. Graphite contains no ions.

QUICK CHECK

The H–O–H bond angle in water is 104.5°, smaller than 109.5°. Why?

- Oxygen is more electronegative than hydrogen.
- Water forms hydrogen bonds.
- Water has only two bonding pairs, which repel each other less.
- The two lone pairs on oxygen repel more strongly than the bonding pairs and push the O–H bonds together.

Oxygen has four electron pairs, arranged roughly tetrahedrally. Lone pairs repel more than bonding pairs, and two lone pairs squeeze the angle by about 5° from 109.5°. Electronegativity affects the polarity of the bonds, not the angle.

QUICK CHECK

In the ammonium ion, NH_4^+, where do the two electrons of the dative bond come from?

- One from nitrogen and one from the hydrogen ion
- Both from the lone pair on nitrogen
- Both from the hydrogen ion
- From the other three N–H bonds

An H^+ ion has no electrons at all, so both electrons in the new bond come from nitrogen's lone pair. Once formed, the four N–H bonds are identical and the ion is tetrahedral.

QUICK CHECK

A grey solid melts at 1083 °C, conducts electricity as a solid and when molten, and is insoluble in water. What is its structure?

- giant ionic
- giant covalent
- giant metallic
- simple molecular

Conducting as a solid points to delocalised electrons, and conducting when molten as well, with a high melting temperature, fits a metal (this is copper). An ionic solid does not conduct until it melts.